Technical Test Submission Report

Fullstack Menu Tree System — Solusi Teknologi Kreatif (STK)

Solusi Teknologi Kreatif

CANDIDATE NAME	APPLIED POSITION	DATE OF SUBMISSION	REPOSITORY VISIBILITY
Luthfi Mirza Darsono	Fullstack Web Developer	23 Oktober 2026	Public Repository

GITHUB REPOSITORY LINK

https://github.com/LuthfiMirza/fullstack-menu-tree-app

Executive Summary & Architectural Architecture

A production-grade fullstack web application implementing a recursive, unlimited-depth hierarchical menu tree with interactive CRUD operations, drag-and-drop reordering & reparenting, real-time search filtering with ancestor preservation, and pixel-perfect Figma UI fidelity.

1. Backend Core & Bonus Features (NestJS 10, TypeORM, PostgreSQL 16)

- ✓ **Adjacency List Pattern:** Parent-child hierarchy with `parentId` foreign key and `ON DELETE CASCADE`.
- ✓ **Derived Dynamic Depth:** Depth is computed on the fly on read via ancestor traversal to prevent denormalization errors.
- ✓ **Transactional Compaction:** Sibling moves and reordering execute inside atomic transactions, guaranteeing gap-free contiguous `0..k−1` positions.
- ✓ **Server-Side Cycle Detection:** Graph traversal algorithm prevents reparenting any node into itself or its own descendants (`409 Conflict`).
- ✓ **OpenAPI / Swagger Documentation:** Interactive API documentation active and live at `/api/docs`.
- ✓ **Health Check Endpoint:** Dedicated `/api/health` endpoint with server uptime and status.

2. Frontend Core & Bonus Features (Next.js 14 App Router, Tailwind CSS, Zustand)

- ✓ **Figma UI Fidelity:** Strict adherence to design tokens (Primary `#0052AF`, Text `#1D2939`, Background `#F8FAFC`).
- ✓ **Unlimited Nested Hierarchy:** Recursive rendering with connector guide lines, toggle chevrons, and Expand/Collapse All controls.
- ✓ **Smart Real-Time Search:** Case-insensitive search that auto-expands all ancestor branches and highlights matching text in yellow.
- ✓ **Interactive Detail Form:** Pinned System Code UUID, Menu ID, Depth, Parent Data, and editable Menu Name with save states.
- ✓ **Smooth Drag-and-Drop (@dnd-kit):** Reorder siblings and reparent nodes across levels with optimistic updates and automatic rollback.
- ✓ **Safe Deletion Dialog:** Confirmation modal showing the exact count of affected sub-menu descendants before cascade removal.
- ✓ **Mobile Responsive Shell:** Sticky brand header with slide-over drawer navigation for mobile and tablet screens.

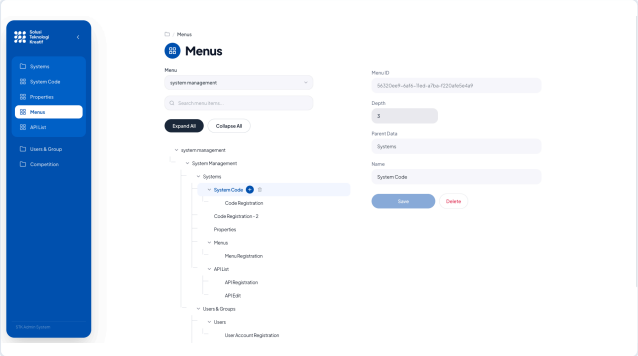
3. Bonus DevOps & Containerization (Docker & Docker Compose)

- ✓ **Development Mode (`docker-compose.dev.yml`):** Live hot-reloading with bind mounts for both frontend and backend.
- ✓ **Production Mode (`docker-compose.yml`):** Optimized multi-stage Docker builds, non-root users, and automated DB migrations.

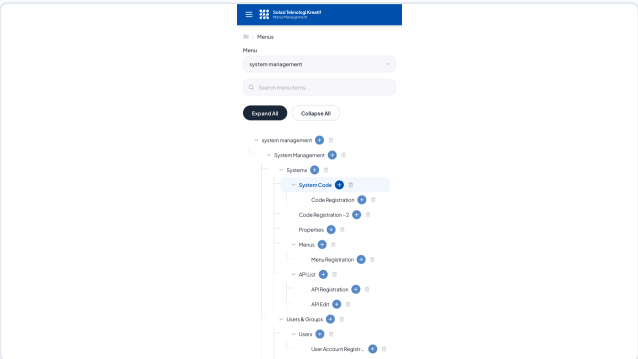
Application Visual Verification (Figma vs Live Implementation)

The interface reflects the Figma specification across key states: desktop hierarchy with detail form, mobile responsive navigation, real-time search filtering, and safe modal confirmations.

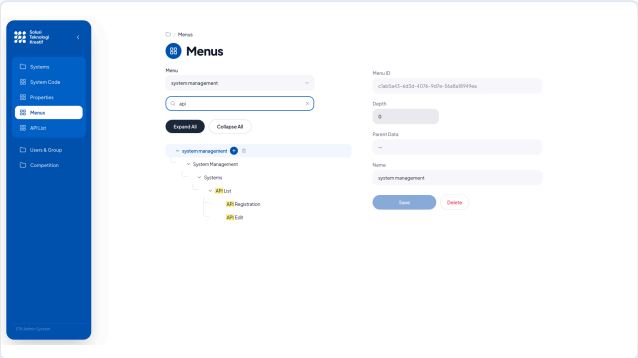
Desktop View (Figma-Aligned Hierarchy & Form)



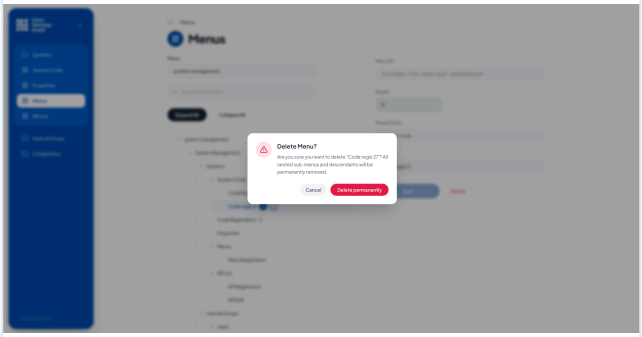
Mobile Responsive Navigation (Drawer & Sticky Header)



Live Search (Yellow Highlight & Ancestor Auto-Expansion)



Safe Cascade Deletion Confirmation Dialog



Automated Unit Tests & Verification

Backend Unit Tests (Jest)

36 / 36 PASSED

Verifies tree construction, cycle detection guard, dynamic depth computation, orphan handling, and transactional move/reorder rollbacks.

Frontend Unit Tests (Vitest)

22 / 22 PASSED

Verifies hierarchical search filtering, ancestor auto-expansion, case-insensitive matchers, tree flattening, and client-side drop guards.

RESTful API Endpoints Summary

Method	Endpoint	Description	Status Code
GET	/api/health	Health status check and system uptime	200 OK
GET	/api/menus	Get full menu tree ordered by contiguous positions	200 OK
GET	/api/menus/:id	Get single menu detail (with derived depth & parentName)	200 OK / 404
POST	/api/menus	Create new root menu or child node under parent	201 Created
PUT	/api/menus/:id	Update menu item name	200 OK / 404
DELETE	/api/menus/:id	Cascade delete menu item and all descendants	204 No Content
PATCH	/api/menus/:id/move	Move menu item to new parent and position (cycle guarded)	200 OK / 409
PATCH	/api/menus/:id/reorder	Reorder menu position within same sibling level	200 OK

Quick Start Execution Guide

Option A — Run with Docker Compose (Recommended):

```
git clone https://github.com/LuthfiMirza/fullstack-menu-tree-app.git cd fullstack-menu-tree-app cp .env.example .env docker compose up --build
```

• Frontend UI: **http://localhost:3000** | • Backend API: **http://localhost:3001/api** | • Swagger Docs: **http://localhost:3001/api/docs**

Option B — Run Locally with Node.js & PostgreSQL:

```
# Backend (Terminal 1) cd backend && cp .env.example .env && npm install npm run migration:run && npm run seed && npm run start:dev # Frontend (Terminal 2) cd frontend && cp .env.example .env.local && npm install && npm run dev
```

Key Architectural Decisions & Edge Case Safeguards

- ✓ **No Hardcoded Depths:** Depth is derived dynamically on read traversal. Moving a subtree with 100 descendants never requires updating 100 depth columns.
- ✓ **Contiguous Position Compaction:** Deleting or moving nodes automatically shifts remaining sibling positions within a single SQL transaction.
- ✓ **Dual-Layer Cycle Guard:** Reparenting checks both client-side and server-side graph descendants to guarantee no circular dependencies can corrupt the tree.